

SANAH A. BHIMANI

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Postdoctoral Research Associate ◊ University of Chicago ◊ Chicago, IL 60622

AREAS OF RESEARCH

I am a cosmologist specializing in millimeter-wave (mm-wave) measurements, where I lead aspects of instrument development, automation, and systematics characterization for ground-based Cosmic Microwave Background (CMB) experiments like the Simons Observatory (SO). I also extend this work to high-frequency mm-wave instrumentation and metamaterials anti-reflection (AR) optical coatings for line-intensity mapping (LIM) experiments.

EDUCATION

Yale University, New Haven, CT; *Advisor: Laura Newburgh*

Aug 2024

Doctor of Philosophy in Physics

Jun 2023

Master of Philosophy in Physics

Dec 2020

Master of Science in Physics

University of California Berkeley, Berkeley, CA

Dec 2017

Bachelor of Arts in Physics & Astrophysics

APPOINTMENTS

Sept 2024–Present

University of Chicago, Chicago, IL
Postdoctoral Research Associate

Jul 2018 – Jul 2019

Yale University, New Haven, CT
Post-baccalaureate Research Assistant

Dec 2017 – Jun 2018

University of California Berkeley, Berkeley, CA
Post-baccalaureate Research Assistant

RESEARCH EXPERIENCE

General AI / ML Applications for Millimeter-Wave Astrophysics

Sept 2024 - present

- Lead software dev for telescope optics production by combining microscope imaging, supervised datasets, U-Net segmentation, and regression modeling to streamline fabrication precision-machined optical components for CMB and LIM experiments
- Co-developing AI-enabled closed-loop manufacturing framework for metamaterial optics that integrates multimodal sensor data, adaptive calibration, and digital-twin models to guide autonomous machining decisions
- Developing unsupervised hybrid anomaly-detection pipeline for SO detector timestreams, using Deep Embedded Clustering and readout-informed detector groupings to distinguish detector systematics from astrophysical transients
- Co-developing agentic frameworks for real-time detector recovery in SO by integrating anomaly-detection outputs, observatory metadata, operator feedback, and guardrailed corrective-action logic to support autonomous instrument optimization

General Millimeter-Wave Optical Instrumentation & Software

Sept 2024 - present

- Developing precision fabrication methods for 850GHz AR-coated optics to advance AR coating technology to the sub-mm wavelength regime for next-gen LIM experiments (e.g., CCAT)
- Fabricating the first broadband metamaterial AR-coated silicon lenses for CCAT/FYST EoR-Spec, supporting 210–420 GHz [CII] line-intensity mapping across the Epoch of Reionization.
- Constructed `mml-automation1`, the first Python-based control system for the Metamaterials AR-Coating Lab, enabling direct Aerotech hardware command removing dependence on proprietary vendor code
- Refactored `SawPy`, a 10+ year legacy codebase, into `MetaLens`—a modular software package that streamlines fabrication workflows by organizing them by optical component type, material-specific process, and calibration method
- Fabricated the first 350 GHz AR-coated alumina filter for CCAT, validating AR-coating performance on alumina at frequencies beyond CMB bands

Simons Observatory (SO)

Jul 2018 - present

- Developing ML pipelines for time-domain anomaly detection to classify detector systematics and enhance sensitivity to astrophysical transients in timestreams
- Fabricated 220-280 GHz AR-coated lenses and filters for the 2026 Advanced-SO deployment to enable CMB high-frequency measurements
- Developed a PCA-based clustering algorithm to identify sky-coupled detectors across all four telescopes to standardize Observatory-wide data-quality cuts / calibration.
- Served on the commissioning team in Chile, bringing SOs distributed data-acquisition and telescope-control software online to support first-light operations for all telescopes.
- Developed novel software technology for data acquisition and telescope control `socs`—focused on cryogenic instrumentation—to ensure stable Observatory operations
- Developed SQL database framework centralizing non-detector metadata (cryogenics, calibration, environment, platform health) across telescope and site-wide systems.
- Led construction, calibration, and analysis of a 280-sensor cryogenic thermometry system, achieving ± 7 mK accuracy; enabled precise detector monitoring and stable cryogenic performance

Hydrogen Probe of the Epoch of Reionization (HYPERION)

2016 - 2018

- Designed and constructed absorber baffles for HYPERION's 30-150 MHz band, comparing pyramidal Zotefoam geometries vs ferrite tiles; performed in-lab and on-site characterization at Owens Valley Radio Observatory with antenna return-loss (S11) measurements
- Applied PCA of simulated Epoch of Reionization signals to classify HYPERION instrument noise

Analyzing Stellar Magnetic Activity via *Kepler* Space Mission

2016 - 2018

- Developed an IDL algorithm combining autocorrelation and periodogram techniques to derive rotation periods of low-mass (cool, low-amplitude) stars from Kepler photometry

Calibrating SNO Detector Response

2015 - 2016

- Constrained changes in angular response for the Sudbury Neutrino Observatory (SNO) using MiniSim, a Monte Carlo program in C++ to help inform calibration methods in support of next-generation experiment SNO+

PRESENTATIONS, PUBLICATIONS & WORKSHOPS

Presentations & Talks

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| June 2024 | <i>The Simons Observatory: Deployment and Current Configuration of the Observatory Control System for SAT-MF1 and Data Access Software Systems.</i> Poster presented at SPIE Astronomical Telescopes + Instrumentation, Yokohama, Japan. |
| January 2019 | <i>Thermometry Integration and Calibration for the Simons Observatory.</i> Poster presented at the 234th Meeting of the American Astronomical Society, Seattle Convention Center. |
| October 2018 | <i>Thermometry Construction and Validation Process for the Simons Observatory.</i> Poster presented at the Wright Laboratory High Energy Physics Instrumentation Seminar, Yale University. |
| 2019 | <i>Cosmology - The Evolution of the Universe.</i> Public talk presented to the local Ismaili Muslim Connecticut community on cosmology, life as an experimental cosmologist, and being a woman in STEM. |
| August 2017 | <i>Stellar Rotation Periods from the Kepler Space Mission.</i> Poster presented at the UC Berkeley Astrophysical Symposium, Berkeley, CA. |
| August 2017 | <i>HYPERION: Characterizing Absorbers at Low Radio Frequencies.</i> Poster presented at the UC Berkeley Astrophysical Symposium, Berkeley, CA. |

Publications with Significant Contributions

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| 2026 | Bhimani, S., Datta, R., McMahon, J., et al. <i>Machine-Learning-Guided Fabrication of Metamaterial Anti-Reflection Coatings for Millimeter-Wave Optics.</i> In preparation. |
| 2025 | Howard, C., Bhimani, S., Datta, R., Thomas, A., et al. <i>CCAT Collaboration: Dual-band optics optimized for the 350 GHz and 405 GHz atmospheric window.</i> Internal Review. |

- 2024 Bhimani, S., et al. *The Simons Observatory: Site Deployment of the Observatory Control System and Data Access Software (SAT-MF1)*. Proc. SPIE 13102,
- 2024 Koopman, B.J., Bhimani, S., Galitzki, N., Hasselfield, M., et al. *The Simons Observatory: Deployment of the Observatory Control System and Supporting Infrastructure*. Proc. SPIE 13102.
- 2024 Nguyen, D., Bhimani, S., Galitzki, N., Koopman, B.J., et al. *The Simons Observatory: Alarms and Detector Quality Monitoring*. Proc. SPIE 13102.
- 2024 Guan, Y., Harrington, K., Lashner, J., Bhimani, S., Crowley, K.T., Galitzki, N., et al. *The Simons Observatory: Observatory Scheduler and Automated Data Processing*. Proc. SPIE 13102
- 2020 Koopman, B.J., Lashner, J., Saunders, L.J., Hasselfield, M., Bhandarkar, T., Bhimani, S. et al. *The Simons Observatory: Overview of Data Acquisition, Control, Monitoring, and Computer Infrastructure*. Proc. SPIE 11452,

Other Publications

- 2026 Thomas, A. et al. *n-Band Scattering and Absorption of Infrared Blocking Foam Filters for Millimeter-Wave Cameras*. Internal review.
- 2025 Simons Observatory Collaboration. *The Simons Observatory: forecasted constraints on primordial gravitational waves with the expanded array of Small Aperture Telescopes* JCAP, arXiv:2512.15833
- 2024 Mangu, A., Corbett, L., Bhimani, S., et al. *The Simons Observatory: Design, Integration, and Current Status of Small Aperture Telescopes*. ApJS, arXiv:2412.01200
- 2024 Galitzki, N., et al. *The Simons Observatory: Design, Integration, and Testing of the Small Aperture Telescopes*. ApJS, arXiv:2405.05550
- 2024 Errard, J., et al. *A Brief Overview of the Simons Observatory*. mmUniverse Proceedings
- 2023 Polzin, A., Asali, Y., Bhimani, S., et al. *Astronomy as a Field: A Guide for Aspiring Astrophysicists*. arXiv:2312.04041
- 2022 Bhandarkar, T., et al. *The Simons Observatory: Development and Validation of the Large Aperture Telescope Receiver*. Proc. SPIE 12190, arXiv:2207.14212
- 2022 CMB-S4 Collaboration, *Snowmass 2021 CMB-S4 White Paper*, arXiv:2203.08024
- 2021 Zhu, N., et al. *The Simons Observatory Large Aperture Telescope Receiver*. ApJS, arXiv:2103.02747
- 2019 CMB-S4 Collaboration, et al. *CMB-S4 Science Case, Reference Design, and Project Plan*. FNAL Technical Report, arXiv:1907.04473
- 2019 Simons Observatory Collaboration, et al. *The Simons Observatory: Astro2020 Decadal Project White Paper*, NASA Technical Report, arXiv:1907.08284
- 2019 CMB-S4 Collaboration, *CMB-S4 Decadal Survey APC White Paper*, NASA Technical Report, arXiv:1908.01062

HONORS & AWARDS

- 2022 **PEO State Scholar Candidate**
Gained Connecticut state recognition by the Philanthropic Educational Organization for demonstrating potential to make significant contributions in the chosen field of study as a woman in a doctoral program

2021 – 2022	Yale Physics Graduate Diversity Fellowship Awarded for implementing a framework for a long-term climate survey and database analyzed by social scientists to support tangible diversity and belonging efforts in the Physics Department
2020 – 2021	Warren A. and Hibernia S. Tyrrell Fellowship Awarded to an outstanding student pursuing a Ph.D. in Physics
2014	Chapin-Herron Scholarship Merit-based scholarship awarded to an undergraduate student at the University of California, Berkeley
2014	Estha M. Rodkey Scholarship Merit- and need-based scholarship awarded to a UC Berkeley undergraduate
2014	Jean C. Witter Jr. Liberty Scholarship Merit-based scholarship awarded to a UC Berkeley undergraduate
2013	Jesse Koshland Scholarship Merit-based scholarship awarded to a UC Berkeley undergraduate
2013	Phoebe A. Hearst Scholarship Merit- and need-based scholarship awarded to a UC Berkeley undergraduate

TEACHING & MENTORING EXPERIENCE

Simons Observatory (SO)

2024 – Present

SO Mentorship Program Mentor

- Provided research and career guidance to early-career graduate student in the Simons Observatory through SOs formal mentorship initiative
- Supported mentee in project planning, technical skill development, networking, and progress across observational and instrumentation work

Independent Research Mentor

Mentor & Research Supervisor

2020 – Present

- Mentored ~90 high school students through physics and astrophysics research projects—ranging from the physics of sailing and rocket propulsion to the Standard Model and CMB cosmology—via Lumiere, Polygence, Indigo Research, Scholar Launch, and Horizon Academics
- Guided ideation, analysis, and manuscript development leading to journal, conference, and showcase submissions
- Delivered individualized mentorship with structured milestones and adaptive support for diverse experience levels

Physics Department, Yale University

Teaching Fellow

2019 – 2024

- Taught undergraduate laboratory and discussion sections for classes ranging from 20 to 150 students across Modern Physical Measurement Lab, Advanced Physics Lab, Advanced Classical Physics, Electromagnetism, and Being Human in STEM
- Guided students through experimental design, data collection and analysis, and scientific writing, emphasizing connections between theoretical concepts and practical measurement
- Provided individualized feedback on lab reports, led office hours, and facilitated review sessions

University of Central Asia, Khorog, Tajikistan

Astronomy Program Developer & Instructor

- 2019 – 2021
- Designed lectures, assignments, and class activities connecting physics concepts with data analysis techniques
 - Developed a summer programming course enabling undergraduates to analyze exoplanet rotation periods using Python and astronomical datasets
- Bay Area Academic Coaching Program** *High School Physics Tutor*
2020 – 2021
- Tutored (remotely) high-school physics students in California through the Aga Khan Development Network education initiative for Afghan refugee students
 - Designed individualized problem sets and conceptual review lectures to strengthen quantitative reasoning
- Yale Pathways to Science: Girls Science Investigation** *Activity Leader & Panelist*
2019 – 2021
- Led hands-on physics demonstrations and discussions for middle school girls to encourage early engagement in STEM
 - Collaborated with faculty and volunteers to design accessible laboratory activities

SERVICE & LEADERSHIP

Simons Observatory (SO)

2020 – Present

Equity, Diversity, and Inclusion (EDI) Lead

- Led collaboration-wide EDI initiatives including organizing town halls and analyzing internal 2020 climate survey
- Partnered with social scientists to develop a robust follow-up survey assessing collaboration climate and inclusivity

2020 – 2022

Ambassador to APS-IDEA Initiative

- Served on the SO committee for the American Physical Society Inclusion, Diversity, and Equity Alliance
- Helped design inclusive collaboration practices and climate assessments for the SO community

2020 – 2022

SO-NSBP Scholars Program Committee Member

- Organized mentor-mentee matching for the inaugural Simons-NSBP Scholars Program, a collaboration between the Simons Foundation and NSBP to advance Black representation in physics
- Provided individualized feedback on graduate school personal statements and developed CMB programming workshops

Yale University

2019 – 2021

Graduate Student Advisory Committee (GSAC) Liason

- Elected to the Yale Physics Department GSAC for two years; served as representative for my cohort and liaison between early-year Physics PhD students and the Director of Graduate Studies

2020

Panelist, Yale Peer Pathways & CuWiP at Yale

- Spoke on graduate school experiences, research paths, and being a woman in physics

Other Initiatives

2017

Mentor, Bay Area Scientists in Schools (BAS)

- Guided 7th- and 8th-grade science projects at Willard Middle School in Berkeley, CA

